PROJECT PHASE - 3

TEAM NO: 58

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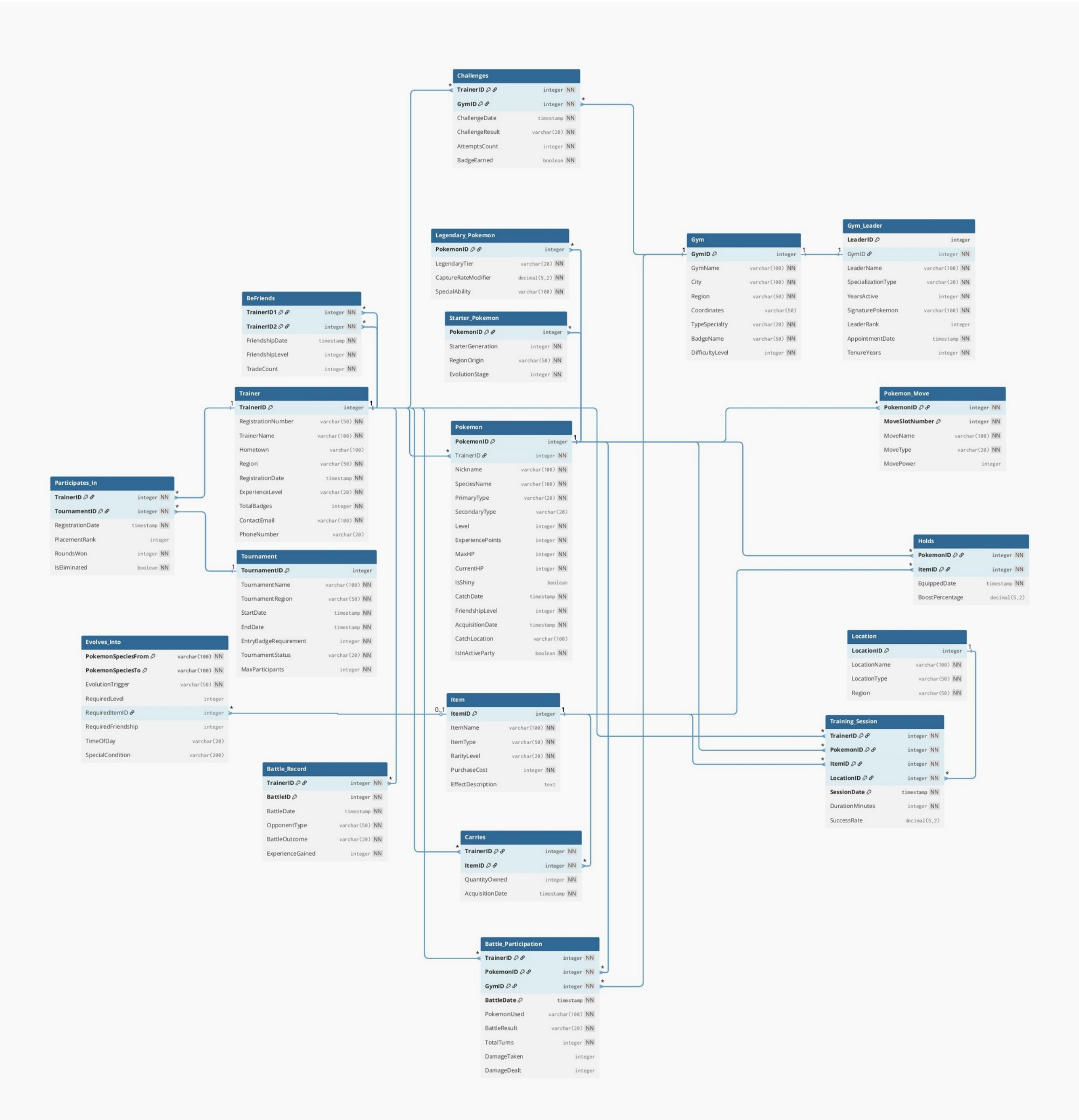
MAPPING ER TO RELATIONAL MODEL

Diagram: ER to Relational Model



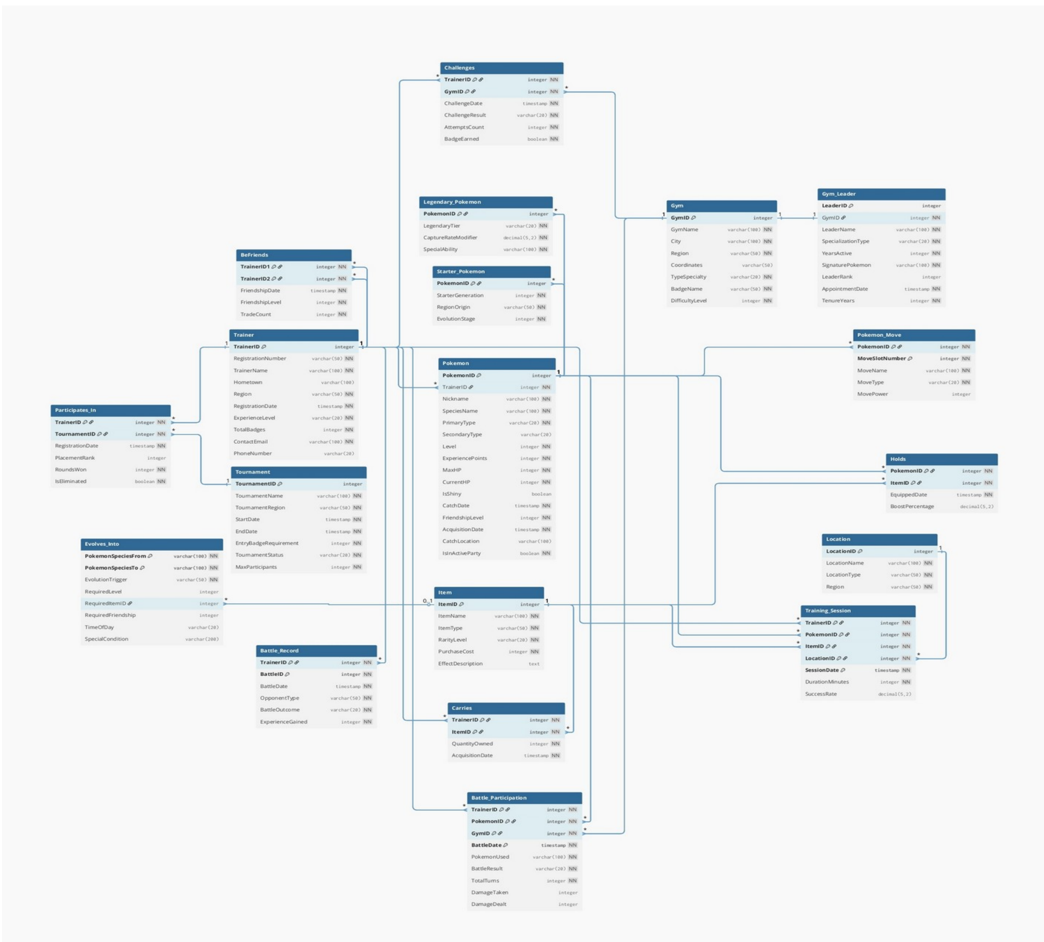
Modifications:

1. We followed the standard ER-to-relational mapping steps from the textbook to convert our ER diagram to the initial relational schema.
2. **Composite Attributes Flattened:**
   - `Trainer.ContactInfo` → `ContactEmail` + `PhoneNumber`
   - `Gym.LocationDetails` → `City` + `Region` + `Coordinates`
   - `Tournament.DateRange` → `StartDate` + `EndDate`
3. **Multi-valued Attributes Converted:**
   - `Pokemon.TypeList` → `PrimaryType` + `SecondaryType` (max 2 types)
   - `Trainer.BadgeCollection` → `Challenges` relationship table
   - `Pokemon.MoveSet` → `Pokemon_Move` weak entity table (max 4 moves)
4. **Weak Entities Created:**
   - `Battle_Record` (depends on Trainer)
   - `Pokemon_Move` (depends on Pokemon)
5. **Relationship Tables Created:**
   - Binary M:N: `Challenges`, `Participates_In`, `Carries`, `BeFriends`, `Holds`, `Evolves_Into`
   - Ternary: `Battle_Participation`
   - Quaternary: `Training_Session`

Total Tables: 19 (7 strong entities, 2 weak entities, 2 specialization tables, 8 relationship tables)

RELATIONAL MODEL TO 1NF

Diagram: 1NF Model same as Relational Model



Modifications:

No new tables added.

The initial relational schema from ER mapping already satisfies 1NF requirements:

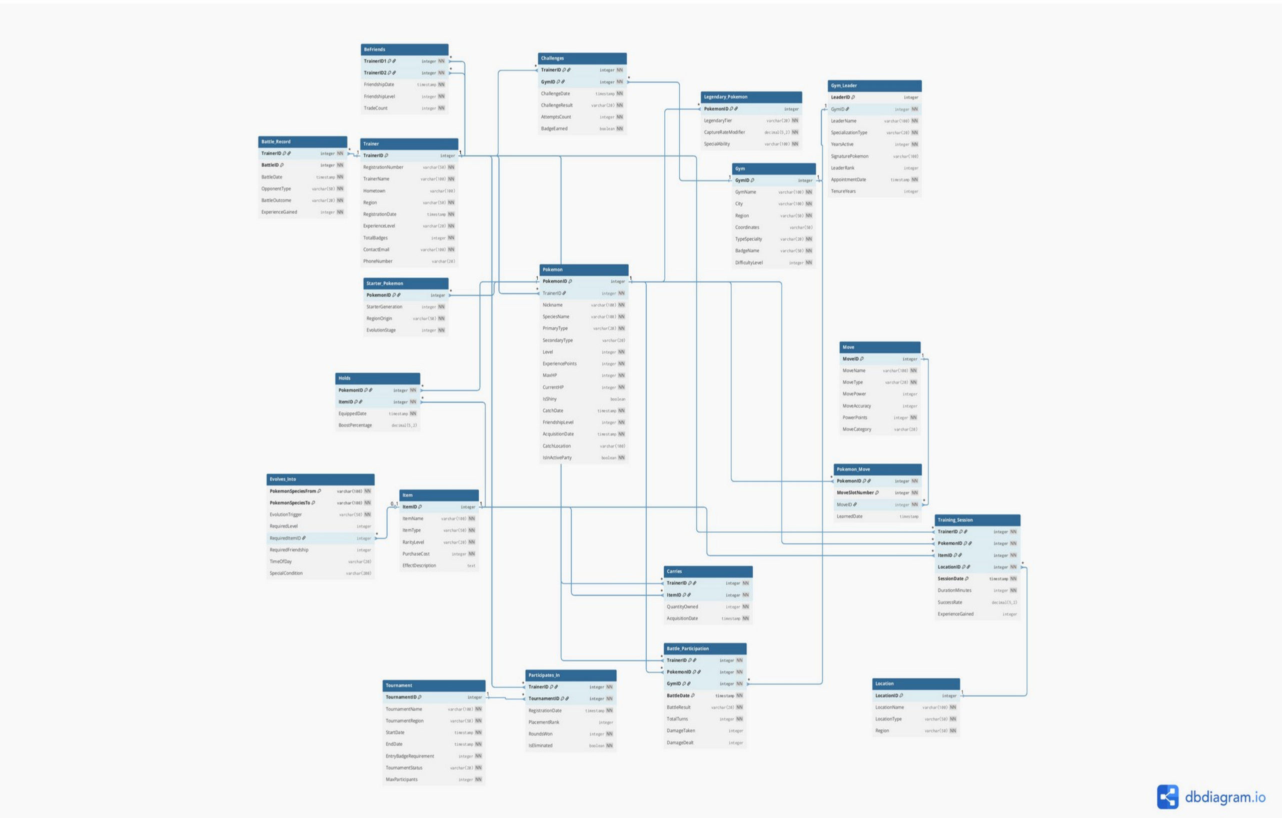
1. **Atomic Values:** All composite attributes already flattened during ER mapping
2. **No Repeating Groups:** All multi-valued attributes already converted to separate tables
3. **Primary Keys Defined:** All 19 tables have proper primary keys
4. **No Duplicate Rows:** Primary keys enforce uniqueness

Verification:

- All attributes contain single atomic values
- No arrays, sets, or nested structures
- Each table has a primary key
- Multi-valued attributes stored in separate tables

1NF TO 2NF

Diagram: 2NF Model



Modifications:

1 New Table Added:

19. MOVE (extracted to eliminate partial dependency)

Changes Made:

Partial Dependency Identified:

- In `Pokemon_Move` table with composite key `(PokemonID, MoveSlotNumber)`:
    - Attributes `MoveName`, `MoveType`, `MovePower` depend only on the move name, not on the full composite key
    - This violates 2NF as non-key attributes depend on part of the primary key

Solution Applied:

1. **Created MOVE table:**
    - `MoveID` (PK), `MoveName`, `MoveType`, `MovePower`, `MoveAccuracy`, `PowerPoints`, `MoveCategory`
2. **Modified Pokemon_Move table:**
    - Removed: `MoveName`, `MoveType`, `MovePower`
    - Added: `MoveID` (FK to MOVE), `LearnedDate`

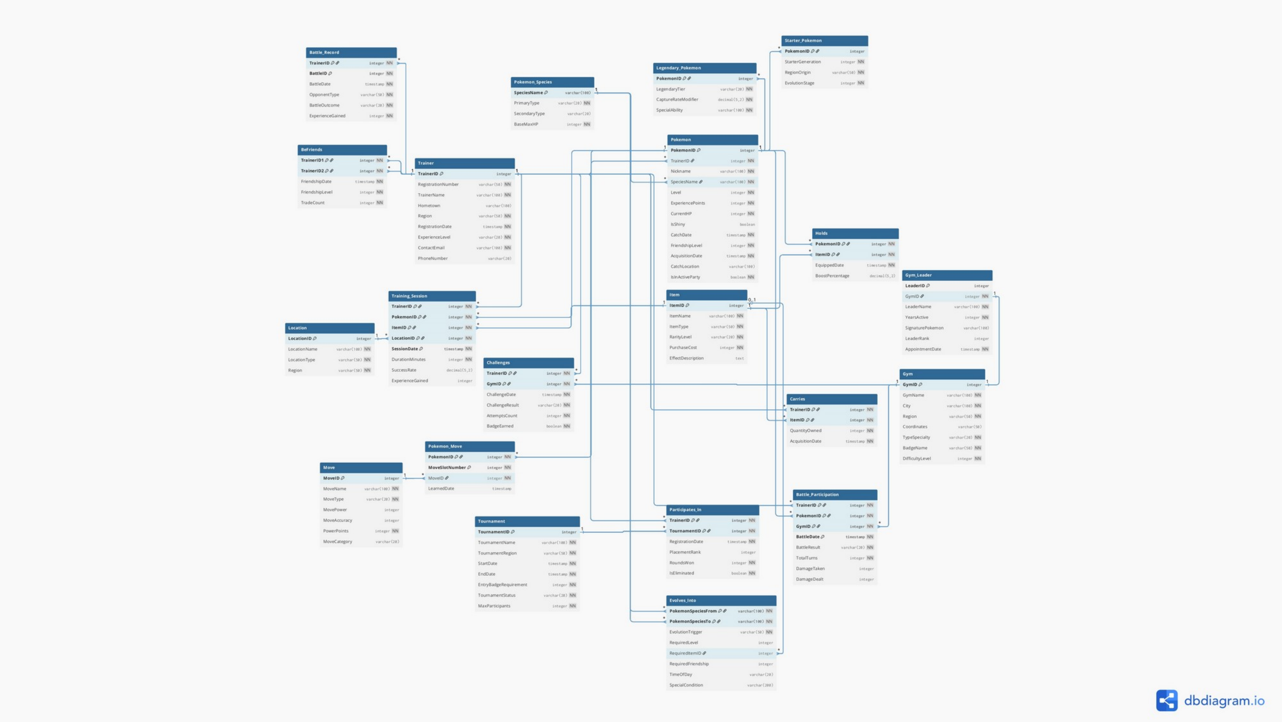
All Other Tables Examined:

- Tables with composite keys: `Challenges`, `Participates_In`, `Carries`, `BeFriends`, `Holds`, `Battle_Participation`, `Training_Session`
- **No other partial dependencies found**
- All non-key attributes depend on complete primary keys

Total Tables: 20

2NF TO 3NF

Diagram: 3NF Model



Modifications:

1 New Table Added:

20. POKEMON_SPECIES (extracted to eliminate transitive dependencies)

Changes Made:

1. Removed Derived Attributes:

- `Trainer.TotalBadges` (can be counted from Challenges table)
- `Gym_Leader.TenureYears` (can be calculated from AppointmentDate)
- `Gym_Leader.SpecializationType` (duplicates Gym.TypeSpecialty)
- `Pokemon.MaxHP` (can be calculated from base stats and level)

2. Created Base Table:

- **POKEMON_SPECIES:** Stores `SpeciesName` (PK), `PrimaryType`, `SecondaryType`, `BaseMaxHP`

3. Fixed Transitive Dependencies:

- **Pokemon table:** Removed `PrimaryType`, `SecondaryType`, `MaxHP`
  - Transitive dependency eliminated: `PokemonID → SpeciesName → Types`
- **Trainer table:** Removed `TotalBadges`
- **Gym_Leader table:** Removed `SpecializationType`, `TenureYears`

4. Verified All Tables:

- All non-key attributes now depend only on the primary key
- No transitive dependencies remain
- No derived attributes stored

Total Tables: 21

There are a total of 4 diagrams in our report:

1. **1st Diagram** – ER to Relational Model (19 tables)
2. **2nd Diagram –** 1NF (19 tables, already satisfied)
3. **3rd Diagram –** 2NF (20 tables, Move table added)
4. **4th Diagram –** 3NF (21 tables, Pokemon_Species added, derived attributes removed)

for clarity

https://drive.google.com/file/d/1wvUOm7U93sBfNPQq5ebEouKhlECPFUAe/view?usp=drivesdk